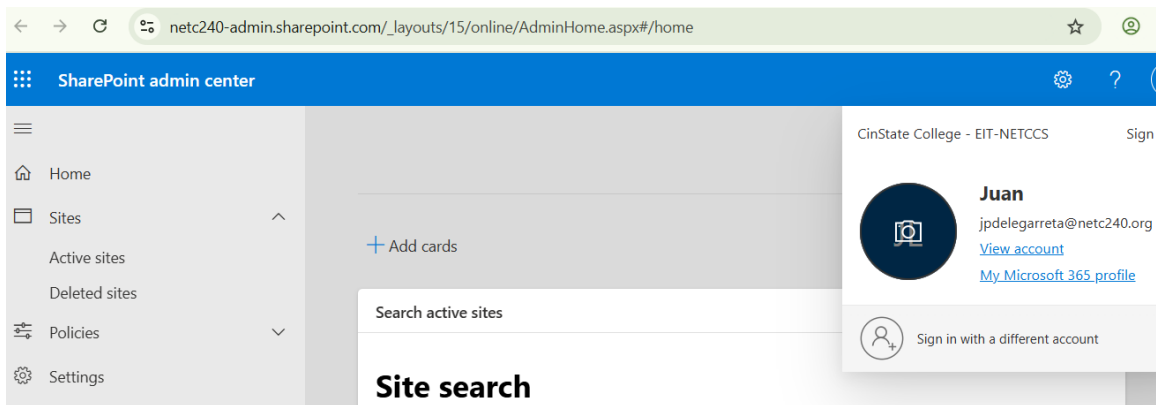


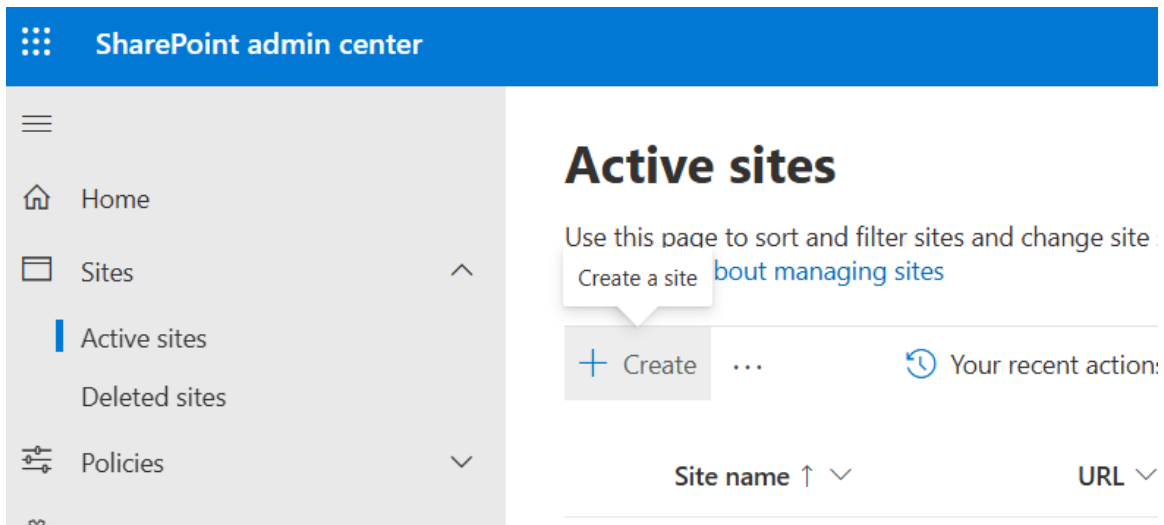
Juan Pablo de Legarreta
NETC-240
Lab 11 – 4-2-2025

Part 1 SharePoint

1. In the SharePoint Admin center create a SharePoint site. Note: this is not the same as creating a site from your specific account.



With my E5 account, I logged into the admin center and clicked on the SharePoint admin center. The screenshot above shows my account logged into the SharePoint admin center.



Under Sites > Active Sites, there is an option to Create a site. I clicked it to begin the process of creating a site.

2. Select the Teams Site option

Create a site: Select the site type

Select the type of site you want to create. [Learn more about team sites](#) or [learn more about communication sites](#).



Team site

Create a private space to collaborate with your team.

- 📅 Track and stay updated on project status
- 🔄 Share team resources and co-author content
- 👥 All site owners and members publish site content
- 🔗 Can connect to other Microsoft 365 products



Communication site

Share information that engages a broad audience.

- 🗂️ Create a portal or subject-focused site
- 👁️ Engage dozens or thousands of viewers
- 👤 Few content authors and many site visitors

I selected the Team site option on the left, seen above.

Select a template

Standard team

Manage projects, share content, and stay connected with your team.

Crisis communication team

Centralize crisis communication, resources, and best practices.

Employee onboarding team

Guide new employees through your team's onboarding process.

Event planning

Coordinate and manage events with your team.

IT help desk

Manage technical requests, track devices, and share training materials.

Project management

Collaborate with your team to share project details and resources.

Retail management team

Unite retail store managers, emphasize store news, and share management resources.

Store collaboration

Coordinate and manage current store and training.

I decided to go with the IT help desk template, inspired by Lab 10.

Give your site a name

Decide on a unique name that follows your organization's naming standards. The description is optional, but useful for people to understand what your site is for.

Your team site will be connected to a Microsoft 365 group, which gives your site a shared OneNote notebook, group email address, and team calendar.



Site name *

de_Legarreta_Lab_11

The site name is available.

Site description

Tell people the purpose of your site

Group email address *

de_Legarreta_Lab_11

The group alias is available.

Site address *

https://netc240.sharepoint.com/sites/de_Legarreta_Lab_11

The site address is available.

Group owner *



Juan

I named the site my last name plus Lab 11 and made my E5 account the group owner.

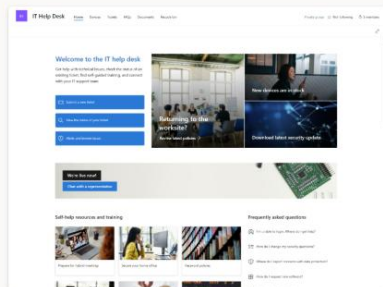
3. The site settings:

a. Private only

b. Only people in your organization. No external sharing allowed

Set language and other options

Set the default language for your site. If required, select labels and other settings to classify what type of content is stored on your site and who should have access to the information.



IT help desk

Change template

Privacy settings

Private - only members can access this site

Select a language

English

Select the default site language for your site. You can't change this later.

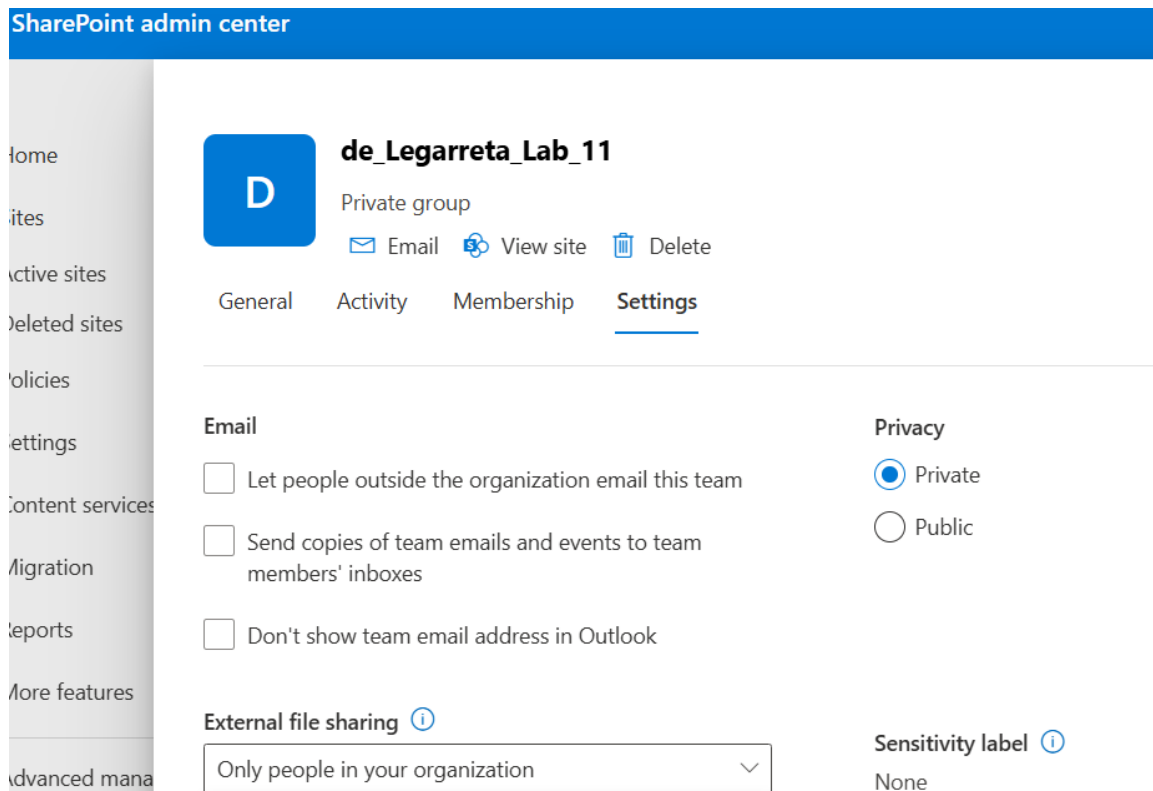
Time zone

(UTC-05:00) Eastern Time (US and Canada)

Back

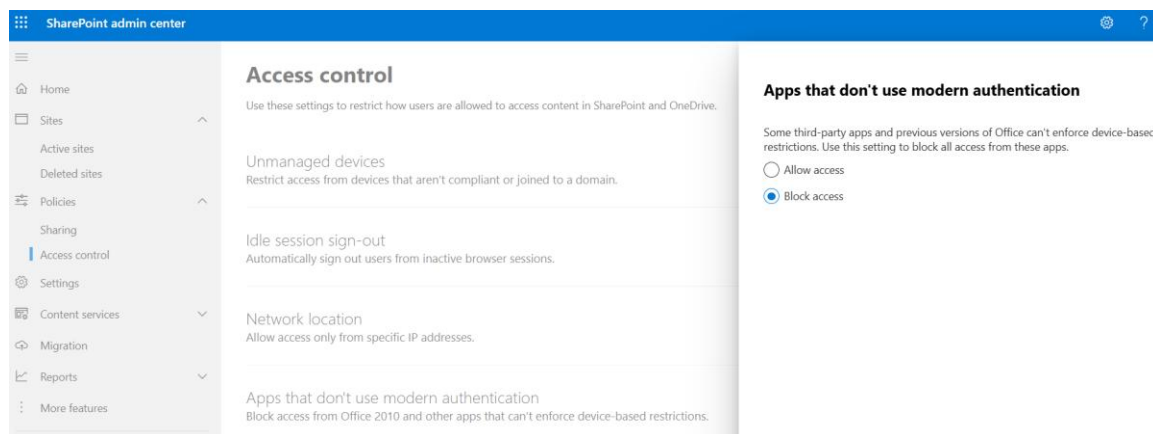
Create site

I set the privacy settings to Private and created the site.



Here I set the External file sharing to ‘Only people in your organization’, as per the lab instructions.

4. Access Control – block apps that don’t use modern authentication



In the SharePoint admin center, you can go to Policies > Access Control > Apps that don’t use modern authentication and toggle the Block Access option, as seen above.

5. Sharing – permission for sharing links should be set to View only

Default sharing link type

Choose the type of link that's selected by default when users share files and folders on this site. [Learn more about the default link types](#)

☒ Same as organization-level setting (Only people in your organization)

- ☐ People with existing access
- ☐ Specific people (only the people the user specifies)
- ☒ Only people in your organization
- ☐ Anyone with the link

Advanced settings for Anyone links ▾

Default link permission

☒ Same as organization-level setting (View)

- ☒ View
- ☐ Edit

Above you can see the default settings are to have links be view-only and limited to only people in my organization.

6. Add your F3 account user



Add group members to de_Legarreta_Lab_11

Members can access everything in the group but can't change group settings. They ca

[Learn more about group permissions](#)

Name



Student6
student6@netc240.org

My F3 account is student 6 – added to the site as a member above.



Add group members to de_Legarreta_Lab_11

✔ Student6 was added as a group member

Members can access everything in the group but can't change group settings. They can also edit the connected SharePoint site.

[Learn more about group permissions](#)

Here is the confirmation that the F3 account was added!

D

de_Legarreta_Lab_11
Private group
 Email View site Delete

General

Activity

Membership

Settings

Owners

Members

Site admins

Site owners

Site members

Site visitors

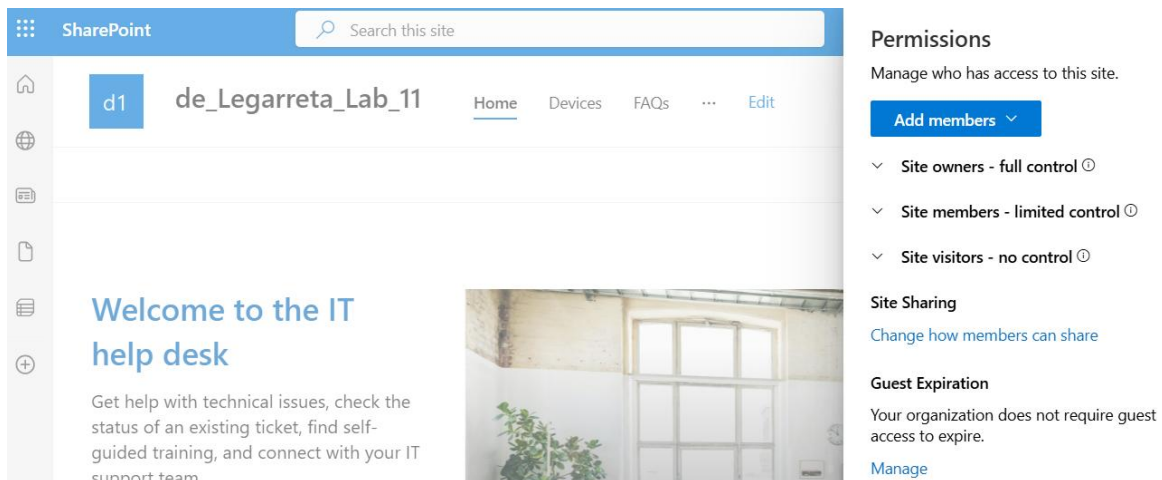
Add members

☐	Name	Email address
☐	Juan	jpdelegarreta@netc240.org
☐	Student6	student6@netc240.org

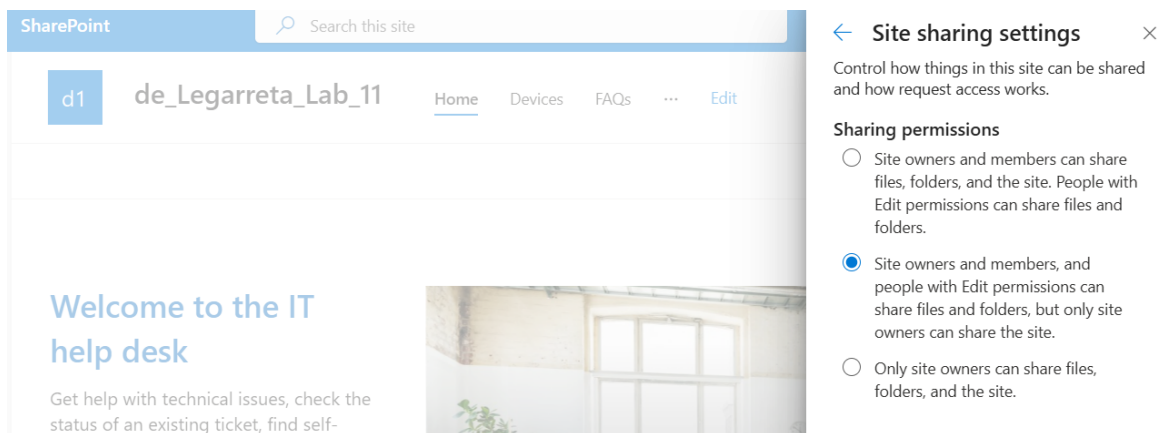
And, sure enough, Student 6 appears under the members list of the group.

7. In Settings (upper right), change the site permissions

- Upload 3 documents to Documents section in the site, with at least one document from your StudentXX account



Now on the SharePoint site, we can change the sharing permissions.



Here, site owners and members can share files and folders, but I changed the setting so only the owner can share the site itself.

8. **The Lab11-files.zip contains 5 files that have filenames that use characters that Windows has no problem with, but SharePoint and OneDrive will not accept.**

Create a program/script, such as PowerShell or Python that will automatically adjust the filenames to remove the prohibited characters. The goal is to leave the filename minimally altered.

Include your script and the output of your script.

Then upload those files to the SharePoint site

```

NETC-240 > file_name_cleanup_script.py > ...
1  import os, re
2
3  # List of the forbidden characters for OneDrive and those found in the files.
4  forbidden_characters = r'[\\/:*?"<>|#&%~]'
5
6  def file_name_cleanup(filename):
7      # I am going to change the forbidden characters into an underscore (_)
8      cleaned_filename = re.sub(forbidden_characters, '_', filename)
9      return cleaned_filename
10
11 def rename_files_in_directory(directory_path):
12     # Get a list of files in the directory
13     files = os.listdir(directory_path)
14
15     for file in files:
16         # Create the full path to the file
17         full_file_path = os.path.join(directory_path, file)
18
19         # Only process files (skip directories)
20         if os.path.isfile(full_file_path):
21             # Clean the filename
22             cleaned_filename = file_name_cleanup(file)
23
24             # If the filename was altered, rename the file
25             if cleaned_filename != file:
26                 # Create the full path for the new file name
27                 new_file_path = os.path.join(directory_path, cleaned_filename)
28
29                 # Rename the file
30                 os.rename(full_file_path, new_file_path)
31                 print(f'Renamed: "{file}" -> "{cleaned_filename}"')
32
33     directory_path = "C:\\Users\\JP\\Downloads\\Lab11-files\\Lab11-files"
34
35     rename_files_in_directory(directory_path)

```

Above is a script that I edited from ChatGPT to rename the files. Note that, though the #, %, &, and ~ characters are not outright banned, I included them in the script to make the files more uniform and compatible.

← Extract Compressed (Zipped) Folders



Select a Destination and Extract Files

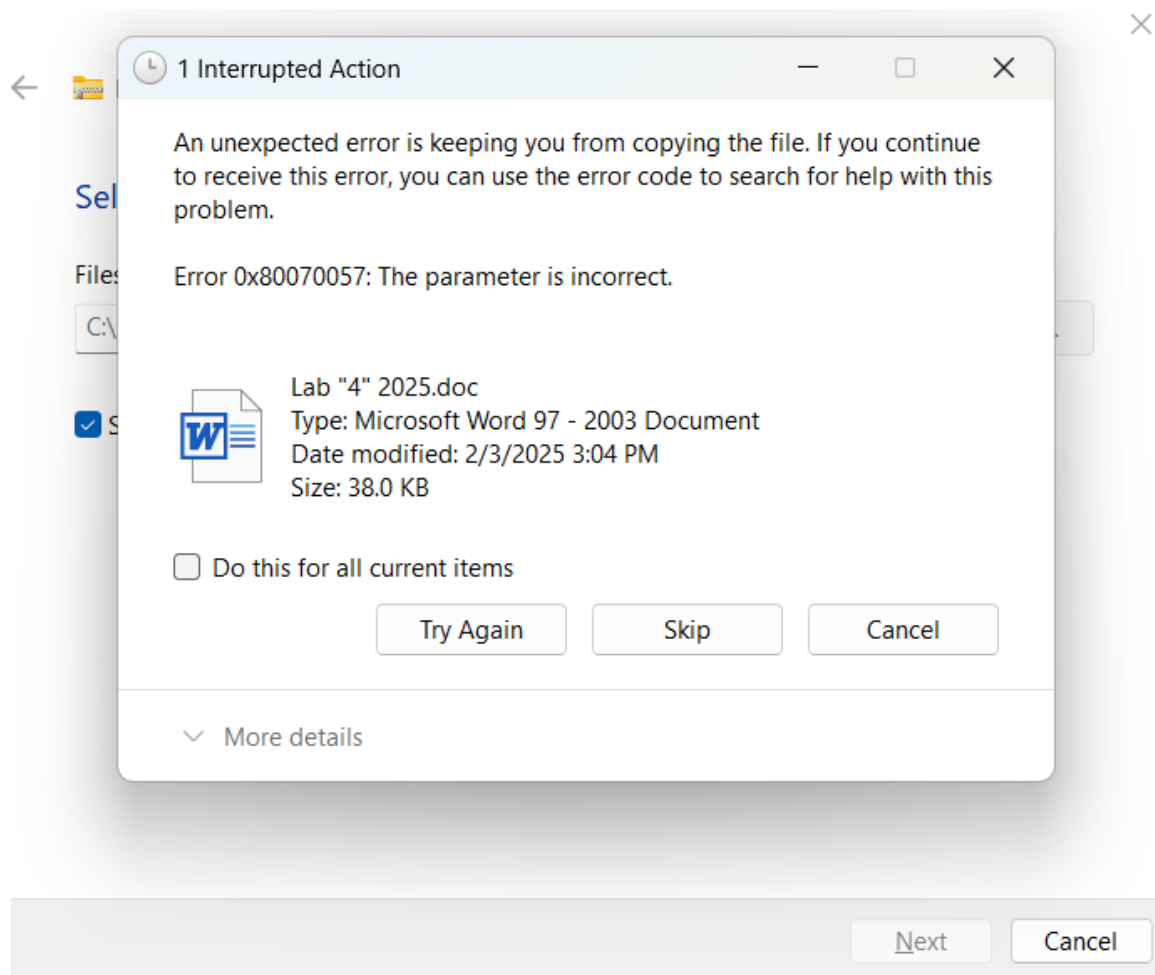
Files will be extracted to this folder:

C:\Users\JP\Downloads\Lab11-files

Browse...

☒ Show extracted files when complete

Before running the script, something curious happened when I extracted the files into my Downloads folder:



It could not copy the file with the name including the “ ” in it. It automatically changed it to underscores (seen below)!

NAME	DATE MODIFIED	TYPE	SIZE
▼ Today			
 Lab #1 1-2025.doc	4/6/2025 9:13 PM	Microsoft Word 97...	115 KB
 Lab &5 2025.doc	4/6/2025 9:13 PM	Microsoft Word 97...	42 KB
 Lab _4_ 2025.doc	4/6/2025 9:12 PM	Microsoft Word 97...	38 KB
 Lab %2 2025.doc	4/6/2025 9:12 PM	Microsoft Word 97...	124 KB
 Lab ~3 2025.doc	4/6/2025 9:12 PM	Microsoft Word 97...	33 KB

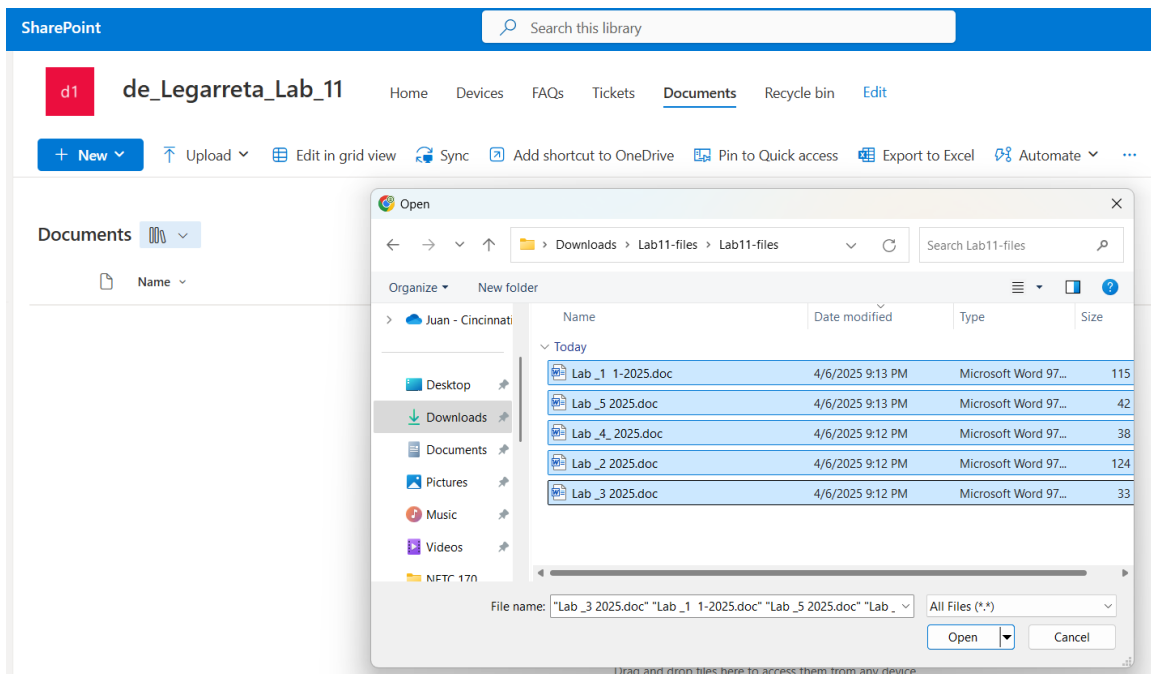
With that in mind, I ran the script anyway.

```
PS C:\Users\JP\Desktop\python_work\netc-240> python file_name_cleanup_script.py
Renamed: "Lab #1 1-2025.doc" -> "Lab _1 1-2025.doc"
Renamed: "Lab %2 2025.doc" -> "Lab _2 2025.doc"
Renamed: "Lab &5 2025.doc" -> "Lab _5 2025.doc"
Renamed: "Lab ~3 2025.doc" -> "Lab _3 2025.doc"
PS C:\Users\JP\Desktop\python_work\netc-240>
```

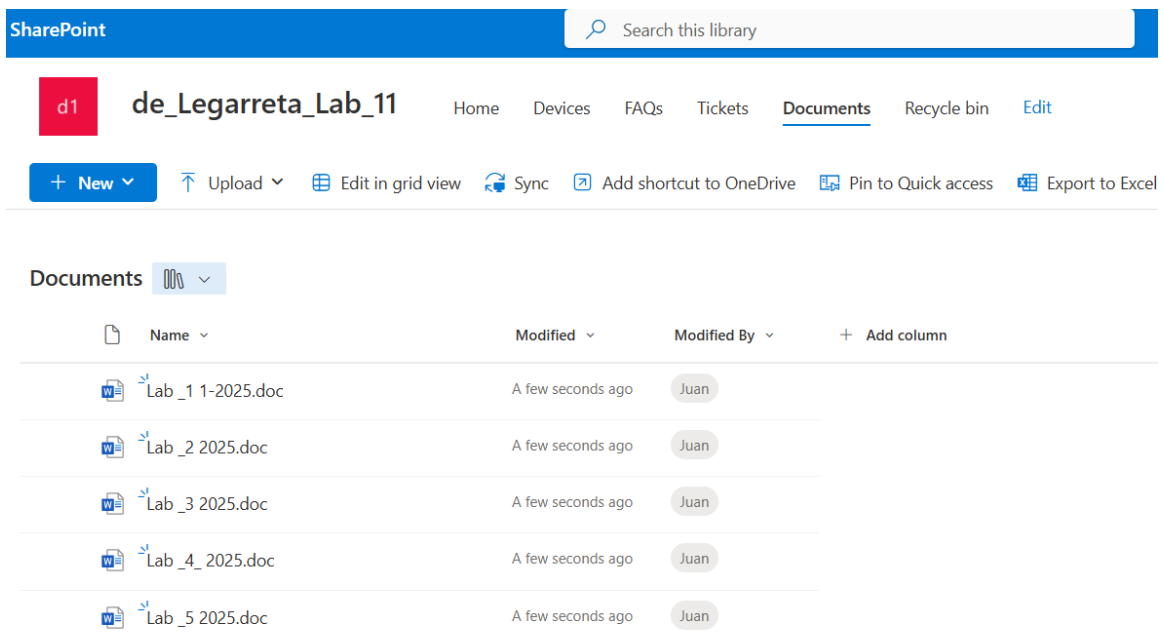
Here is the output of the script. Notice that Lab 4 is not there, as it was already ‘changed’ by the file extraction process.

NAME	DATE MODIFIED	TYPE	SIZE
▼ Today			
 Lab _1 1-2025.doc	4/6/2025 9:13 PM	Microsoft Word 97...	115 KB
 Lab _5 2025.doc	4/6/2025 9:13 PM	Microsoft Word 97...	42 KB
 Lab _4_ 2025.doc	4/6/2025 9:12 PM	Microsoft Word 97...	38 KB
 Lab _2 2025.doc	4/6/2025 9:12 PM	Microsoft Word 97...	124 KB
 Lab _3 2025.doc	4/6/2025 9:12 PM	Microsoft Word 97...	33 KB

You can see the files have changed names now. I will upload them to the SharePoint site I created earlier.



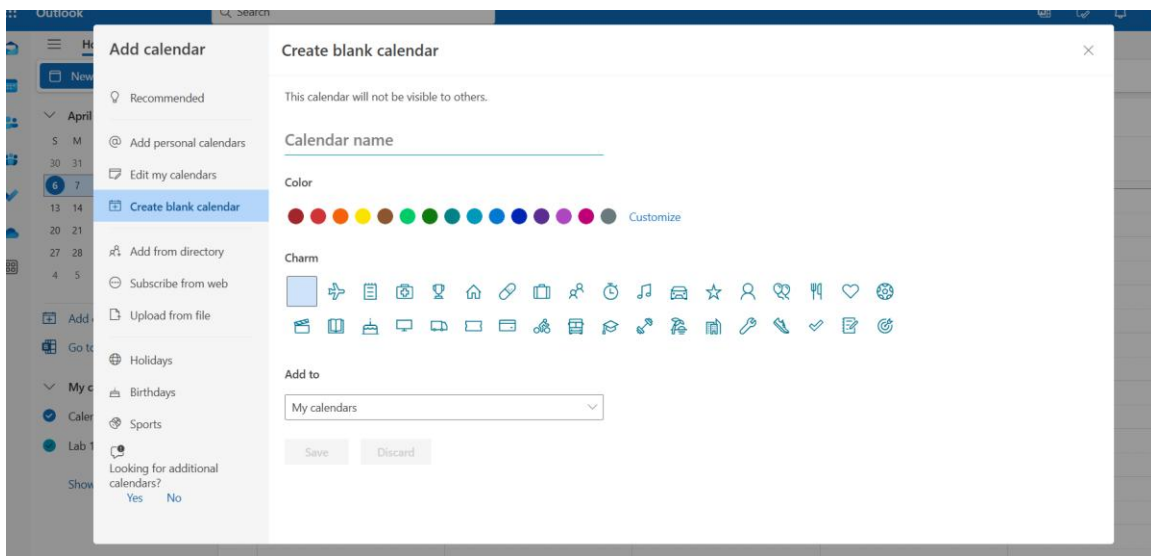
I am uploading the files to the SharePoint Lab 11 site.



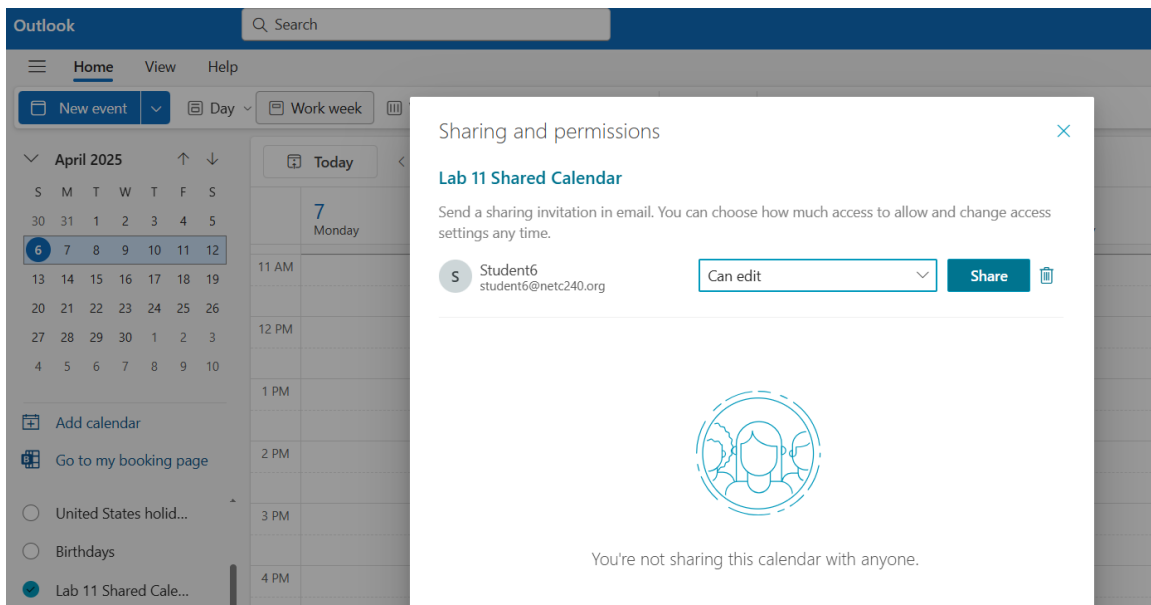
And here they are in the site itself!

Part 2.

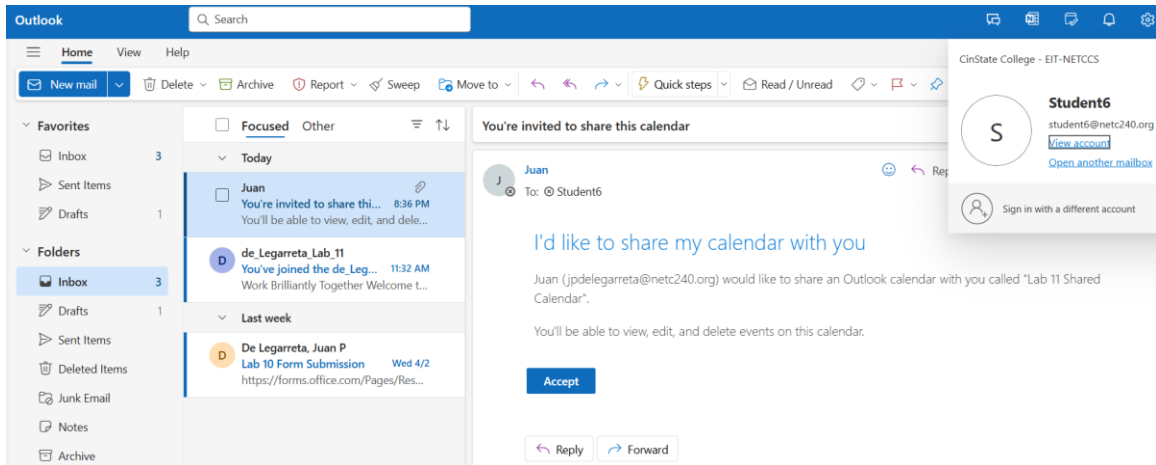
1. Create a Shared Calendar that can be accessed by both your E5 and F3 accounts.
2. Both accounts should have full access to the calendar and all calendar items



In Outlook, I created a blank calendar called Lab 11 Shared Calendar.

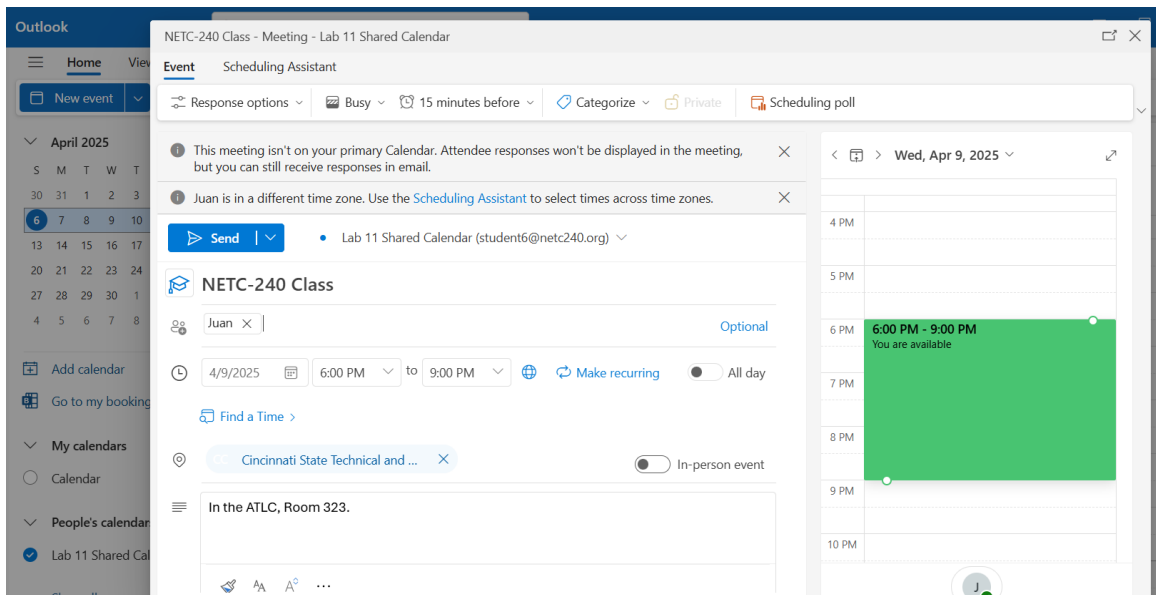


I added the student6 email address and changed the permissions to 'can edit'.

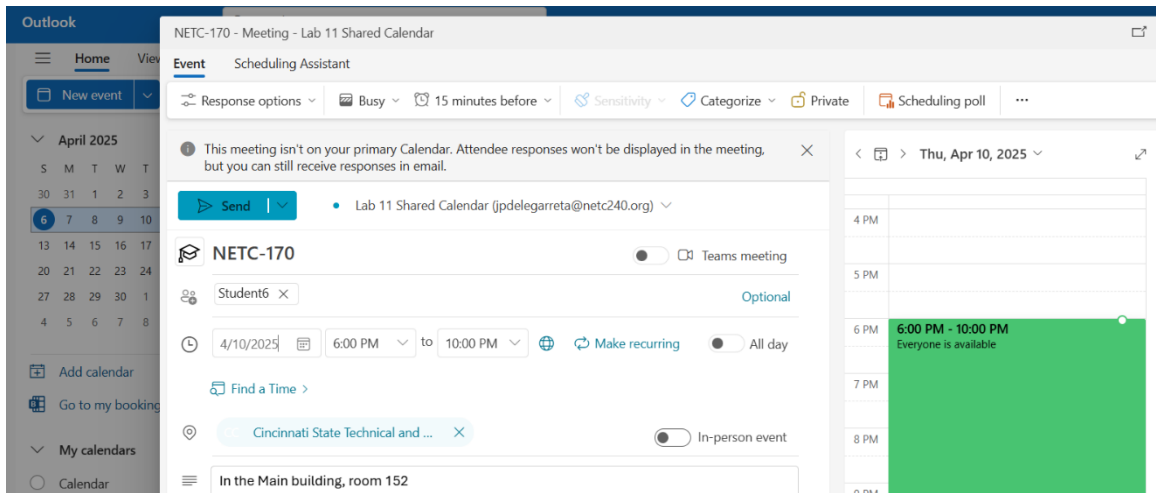


Student6 got an email invitation to access this calendar!

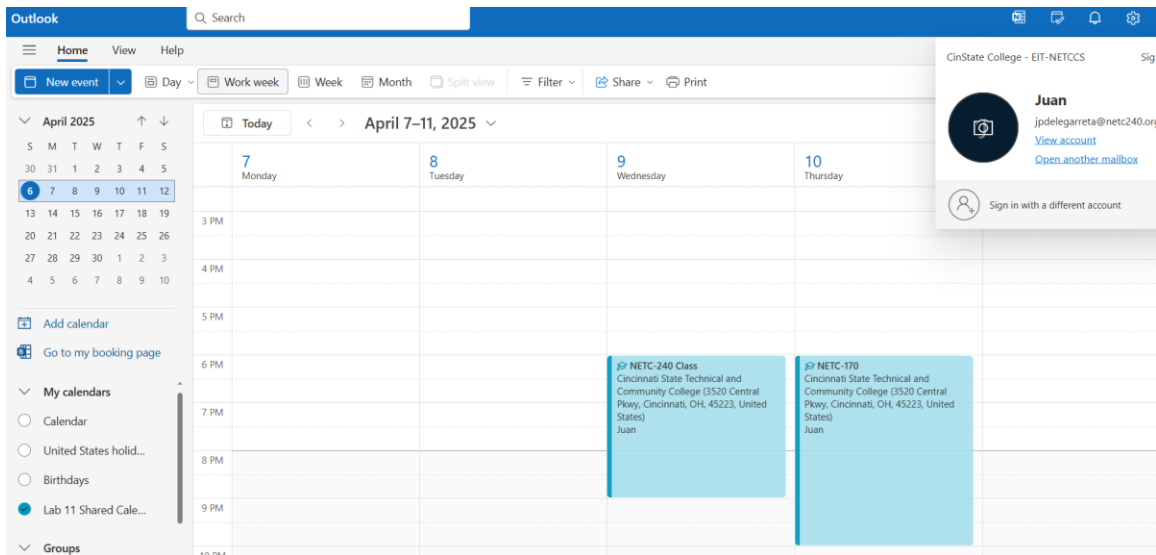
3. Populate the calendar with two events.



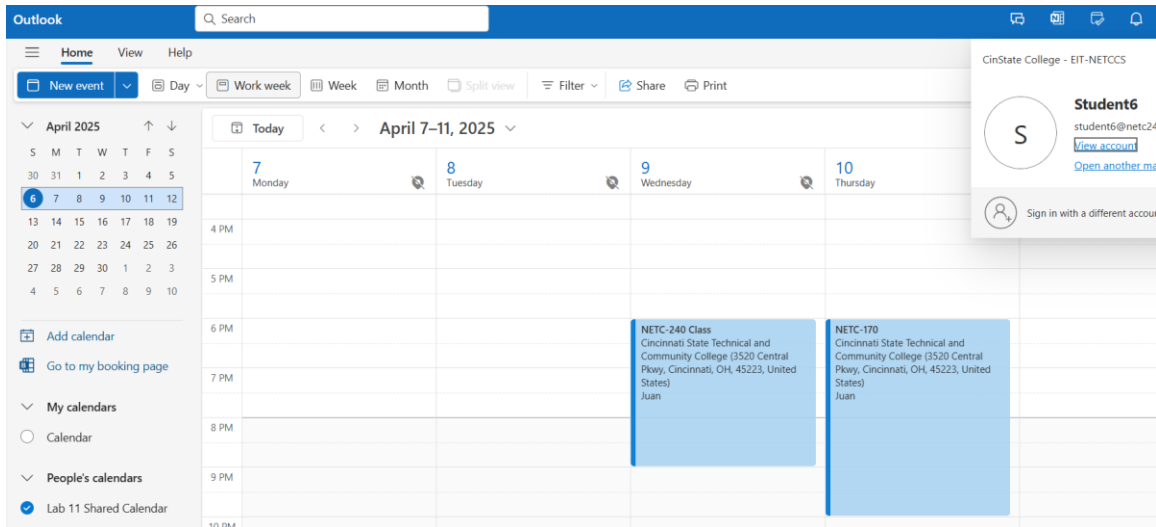
Student 6 is adding an event for the next NETC-240 class. I even invited my E5 account (“Juan”) to the event!



My E5 account is making an event for the NETC 170 class on Thursday night and inviting Student6 (my F3 account).



Here you can see both events in my E5 account (above) and my F3 account (below)!



Questions:

Q1. Create a table that shows the characters which are not allowed. The table should have 3 columns (Char, Description, Reason disallowed)

Forbidden Character	Description	Reason Why It Is Forbidden
\	A backslash used for directory separation in file paths	The backslash is a reserved character used by the operating system to separate directories in file paths. It would cause conflicts in the file system.
/	A forward slash used for directory separation in URLs and file paths	The forward slash serves as a directory separator in many file systems and URLs, so using it in file names would create pathing issues.
:	A colon, often used to specify drive letters or time in filenames	The colon is reserved for specific system functions, such as drive letters (e.g., C:) and time representations. Its use in file names would conflict with these functions.
*	An asterisk used as a wildcard character	The asterisk is used by the system to denote a wildcard, which can represent multiple characters in file searches or command-line operations, making it problematic in filenames.
?	A question mark used to represent an unknown value or wildcard	The question mark is used to represent a single unknown character in file systems or command-line operations, so it's not allowed in filenames.
"	Double quotation marks	Quotation marks are used to enclose file paths and command arguments, so using them in file names would create conflicts when referencing the file.
<	Less-than symbol, used for comparisons	The less-than symbol is reserved for comparisons and operations in programming and file systems. It would interfere with file operations if used in filenames.

>	Greater-than symbol, used for comparisons	Similar to the less-than symbol, the greater-than symbol is reserved for comparisons and file operations, creating potential issues in file management.
	A vertical bar (pipe) used in command-line operations	Allowing the pipe character in file names could interfere with the system's ability to process or interpret file names properly, especially when performing operations or scripts that involve piping output between commands.

Source:

<https://support.microsoft.com/en-us/office/restrictions-and-limitations-in-onedrive-and-sharepoint-64883a5d-228e-48f5-b3d2-eb39e07630fa#invalidcharacters>

<https://stackoverflow.com/questions/1976007/what-characters-are-forbidden-in-windows-and-linux-directory-names>

Q2. Why can it be important to create a SharePoint site at the tenant vs. user level?

Creating tenant-wide SharePoint sites makes it part of the infrastructure of the organization's SharePoint. This makes it easier for administrators to apply policies, restrictions, security, and permissions from the administrative portal. Sites created at the tenant level are also designed for collaboration in mind – able to be accessed by multiple users or groups. Perhaps one of the most important reasons that tenant-level SharePoint sites are preferable to individual SharePoint sites is that the tenant-level sites are not tied to a particular user.

User level SharePoint sites are much more closely tied to a particular user, and the organization does not have as easy access to the permissions, security, and policies associated with an individual user's SharePoint site. In terms of sharing, individual SharePoint sites are better suited for small group collaboration, not organization-wide collaboration. Individual-level sites are tied to a user, which might present issues if a user can't access their account, is hacked, or leaves the organization.

Sources:

<https://learn.microsoft.com/en-us/sharepoint/restricted-site-creation>

Q3. Find the maximum absolute path length in Windows, Mac and Ubuntu Linux. Then explain how those who use a long naming convention can run into problems, when files are embedded several file folders down.

Windows: 260 characters (normal); 32,767 (long path enabled)

Mac: 1024 characters

Ubuntu Linux: 4096 characters

When files are embedded several levels down or use long file names, the problems that can occur include errors in accessing files, issues with compatibility with applications or APIs, problems with synchronization or backups, even issues with scripts. To mitigate this, a file taxonomy should be mindful of not going too many layers deep.

<https://learn.microsoft.com/en-us/windows/win32/fileio/maximum-file-path-limitation?tabs=registry>

<https://stackoverflow.com/questions/7140575/mac-os-x-lion-what-is-the-max-path-length?>

<https://askubuntu.com/questions/859945/what-is-the-maximum-length-of-a-file-path-in-ubuntu?>

Q4. Your organization is preparing to roll out SharePoint across multiple departments. As part of the information governance strategy, you've been asked to develop a naming taxonomy for documents and folders to ensure consistency, searchability, and compliance.

Create a naming convention/taxonomy for documents and folders that will be stored in SharePoint across departments (e.g., HR, IT, Finance). Your taxonomy should account for version control, dates, document type, and department.

Then, evaluate your taxonomy against the following criteria:

- Scalability across new departments
- Ease of use and adoption by staff
- Compatibility with SharePoint and OneDrive restrictions
- Search and filtering efficiency

Include your rationale about the choices you made in your taxonomy design.

A naming convention for **files** could be the following:

Department_Document Type_Date-Version_Descriptive Name

For example, an IT governance audit report, second revision from April 1, 2025 could be:

IT_Audits_2025-04-05_V2_CambridgeAuditReportUpdated

A consultant salary scale for HR, third revision from January 23 2025 could be:

HR_SalaryScale_2025-01-23_V3_ConsultantSalaryScaleLevel3

A naming convention for **folders** could be the following:

Department_Document Type

For example, the IT governance audit report could be in the following folder:

IT_Audits

The salary scale could be in:

HR_SalaryScale

Scalability:

Using this intuitive, simplified but descriptive naming convention makes it easy to scale. It is also flexible – if an organization wanted to add an office location to the beginning of the naming convention, such as:

Toronto_IT_Audits

Ease of use and adoption by staff:

The system is intuitive and easy to understand. It does require a little bit of work, but most people already create descriptive file names. It shouldn't be too difficult to encourage them to add the new parts at the beginning of a file name.

Compatibility with SharePoint/OneDrive restrictions:

Removing the banned characters from the naming guidelines makes them compatible with OneDrive and SharePoint. It would be important to include instructions/policies/norms to keep the 9 banned characters out of the file names. Furthermore, this would hopefully be able to house organization-wide documents within 3 or 4 levels from the very top: an IT SharePoint site with folders that indicate the files within, and then the files themselves. Maybe 1 or 2 more layers of folders for a year or a quarter, or something like that. This helps keep the max path shorter as well.

Search and filtering efficiency:

If each folder is headed by a department, searching for all of the folders in a department would be very easy. Since the second part of the folder is about the files it contains, it should be very easy to find where specific documents would be stored.

File-wise, given that the first two parts of the filename would be shared with its parent folder, they should be easy to find in the folder structure. Furthermore, adding the date and the version number can make searching for older/newer versions easier.